Fiscal Year:	FY 2015	**Task Last Updated:**	FY 09/18/2014
PI Name:	Foster, Jamie Ph.D.		
Project Title:	Impact of Microgravity on the Cell-Cell Interactions between a Mutualistic Bacterium and Its Animal Host		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Merritt Island	**State:**	FL
Zip Code:	32953	**Congressional District:**	3
Comments:	EDITOR'S NOTE: Congressional district is for University of Florida, Gainesville. Address is PI's physical address.		
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	12/01/2013	**End Date:**	11/30/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Sept 2014: As this was the first year of the award a postdoctoral fellow was hired.		
COI Name (Institution):			
Grant/Contract No.:	NNX13AM44G		
Performance Goal No.:			
Performance Goal Text:			

Beneficial microbes play a significant role in the normal development of animal tissues. Through the formation of healthy stable associations between host and symbiont, particularly during post-embryonic development, carefully regulated developmental pathways can be induced.
In this experimental research plan, we propose to examine the impact of microgravity on the normal molecular interactions between host and symbiont. To examine these effects we propose to use the model system between the squid Euprymna scolopes and the luminescent bacterium Vibrio fischeri. For over two decades this squid-vibrio model has served as an ideal ecosystem to examine the influences of mutualistic bacteria on animal development. To simulate microgravity, rotating bioreactors called high-aspect-ratio rotating wall vessel bioreactors (HARVs) will be used to provide a low-shear environment for both the host and symbiont, thereby simulating microgravity. Specifically, we

Task Description:	propose the following: Objective 1. Examine the mechanism(s) associated with the delay of bacteria-induced immune cell trafficking that occurs in simulated microgravity Objective 2. Characterize the changes in the transcriptome of the host squid in the presence and absence of symbiotic bacteria exposure under microgravity conditions. Objective 3. Develop a flight-ready microtiter plate for future flight experiments and optimize the conditions needed to initiate and maintain the symbiosis in this new hardware. The proposed work has several implications for the field of space biology. First, by examining how normal healthy symbioses are initiated and maintained under microgravity conditions, we can begin to determine the sensitivity of these ecosystems to perturbations in the space environment. Secondly, we can assess the genetic and metabolic responses to gravity in the presence and absence of symbiosis, potentially identifying previously unknown triggers or cues associated with post-embryonic animal development.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	All animals form symbiotic associations with bacteria. These ubiquitous interactions are essential for the normal development and health of the host organism. This project helps to improve our understanding of the molecular cross-talk that occurs between animals and microbes by delineating those genes and pathways that are influenced by gravity. As the force of gravity has not changed throughout Earth's history we have a unique opportunity to assess how these host-microbe interactions alter gene expression and their physiology in the absence of gravity.
Task Progress:	Task 1. Develop a flight-ready microtiter plate for future flight experiments and optimize the conditions needed to initiate and maintain the symbiosis in this new hardware. A prototype has been constructed to house six replicate juvenile squid that includes both activation and deactivation chambers. The prototype hardware is estimated to be available by October 1, 2014 for testing with the animals to ensure its effectiveness. Task 2. Characterize the changes in the transcriptome of the host squid in the presence and absence of symbiotic bacteria under microgravity conditions. We have divided this task into two primary projects: A) examine the symbiont transcriptome in the presence or absence of gravity; and B) examine the host transcriptome. In YR1 we have completed the first component. A) Examining the symbiont transcriptome: Cultures of wild-type Vibrio fischeri and a an hfq defective mutant V. fischeri were exposed to simulated microgravity conditions using High-aspect ratio bioreactor chambers. The gene hfq was targeted as previous research on known pathogens has identified this gene as a global regulator that is significantly down-regulated during simulated microgravity and actual space flight conditions. Total RNA was isolated from each treatment and then processed to remove ribosomal RNA. We then generated cDNA libraries to the remaining message RNA and then sequenced the libraries using a NexSeq Illumina platform. There were eight individual treatments and libraries were generated in triplicate for the treatments for a total of 24 cDNA libraries. We have received the sequences and are now completing the bioinformatic analysis and comparison of the libraries to assess which genes were differentially expressed in the symbiotic bacteria under simulated microgravity conditions. We will also assess the impact microgravity had on gene expression on mutants defective in the global regulator hfq. We anticipate differences will be present in the relative abundance of virulence genes. Importance of this task: this work will provide a key base of target genes to explore in greater detail and potentially generate mutants to assess if there is a microgravity phenotype associated with V. fischeri. B) Examining the host transcriptome: A postdoc was hired in July 2014 to begin this task. We have begun the exposure of host animals in the presence and absence of the bacteria and started to accumulate enough tissue for RNA extraction. We are also in the process of optimizing the RNA extraction protocol for the host tissue. Currently, we can extract 1 microgram of total RNA from 10 juvenile light organs (i.e., the site of symbiosis in the host squid). We are trying to improve on this so that we can reach the same concentration with three animals as that the prototype flight hardware will only house six animals per plate. Task 3. Examine the mechanism(s) associated with the delay of bacteria-induced immune cell trafficking that occurs in simulated microgravity. Work on this task is just beginning as a new graduate student was been assigned to this research objective in July of 2014 and is completing her coursework. This task will be a primary focus of YR2.
Bibliography Type:	Description: (Last Updated: 04/20/2026)
Articles in Peer-reviewed Journals	Foster JS, Wheeler RM, Pamphile R. "Host-microbe interactions in microgravity: assessment and implications." Life. 2014;4(2):250-66. Published 26 May 2014. http://dx.doi.org/10.3390/life4020250 , May-2014